



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

PIP PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Build & Package Management

TOTAL: 10 QUESTIONS

Q1 What is pip and what UI problem does it address?

Threat Model

Q6 How do you fetch data and handle loading/error states in pip?

Q2 How do components communicate in pip?

Architecture

Q7 What performance techniques apply to pip applications?

Lifecycle

Q3 How does pip manage state for complex applications?

Configuration

Q8 How do you handle forms and validation in pip?

Compliance

Q4 What rendering strategies does pip support?

Hardening

Q9 How do you test pip components?

Testing

Q5 How does pip handle client-side routing?

SSO / IdP

Q10 What accessibility (a11y) practices are essential in pip?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 pip is a component-based library or framework

Mitigation

pip is a component-based library or framework for building interactive user interfaces. It provides a reactive programming model where UI updates automatically when underlying state changes, reducing imperative DOM manipulation.

A6 Use a data fetching hook or library

Observability

Use a data fetching hook or library that returns { data, isLoading, error }. Show a skeleton or spinner while loading, an error message on failure, and the data on success. Avoid fetching in render to prevent waterfalls.

A2 Parent-to-child communication uses props (one-way data flow)

Minimizes

Parent-to-child communication uses props (one-way data flow). Child-to-parent uses events or callbacks. Siblings share state via a common parent or a global store (Redux, Pinia, Signals). Avoid prop-drilling by using context or state management.

A7 Code splitting loads only the code needed

Automation

Code splitting loads only the code needed for the current route. Memoization (useMemo, React.memo) prevents unnecessary re-renders. Virtualization renders only visible list items. Images use lazy loading and next-gen formats (WebP, AVIF).

A3 Local component state handles UI-only concerns. Shared state

Hardening

Local component state handles UI-only concerns. Shared state (user auth, cart) lives in a global store with a predictable update pattern (actions, reducers, or mutations). Server state (API data) is managed by libraries like React Query or SWR.

A8 pip manages form state with controlled inputs

Governance

pip manages form state with controlled inputs or a form library (React Hook Form, VeeValidate). Validation runs on blur or submit, displaying field-level error messages. Schema libraries (Zod, Yup) define validation rules declaratively.

A4 pip supports CSR (browser renders), SSR (server renders)

Gotchas

pip supports CSR (browser renders), SSR (server renders HTML on each request), SSG (pages generated at build time), and ISR (pages regenerated on a schedule or on-demand). Choose based on SEO requirements and data freshness needs.

A9 Unit tests check individual component rendering and interactions

Assessment

Unit tests check individual component rendering and interactions using a component testing library. Integration tests verify multi-component flows. End-to-end tests (Playwright, Cypress) simulate real user journeys in a browser.

A5 pip's router (built-in or a library like React Router)

Federation

pip's router (built-in or a library like React Router) maps URL paths to components without page reloads. Dynamic segments (/users/:id) are parsed and passed as params. Guards or middleware can protect routes requiring authentication.

A10 Use semantic HTML (button, nav, main) instead

Optimization

Use semantic HTML (button, nav, main) instead of divs with click handlers. Ensure keyboard navigability (tab order, focus management). Add ARIA roles and labels for dynamic content. Test with a screen reader and run axe or Lighthouse audits.